**Mock for National Examination**
**Paper:**
**Applied Mathematics A**

**Level: L5**

# MOCK FOR THE 2025/2026 NATIONAL EXAMINATIONS

# QUESTION PAPER

| **TRADE:** | ***Software Programming and Embedded System*** |
|---|---|

**CANDIDATE NAMES:** ……

**INDEX NUMBER:** ……

**DURATION:3hours**

**INSTRUCTIONS TO CANDIDATES:**

1. **Read and understand** all instructions before answering.
2. For multiple-choice questions, **choose only one correct answer** and clearly **circle the letter or number**.
3. **Erasing or altering** answers is strictly prohibited and will lead to **zero mark** for the affected question.
4. All answers for **open-Ended questions** must be written in the spaces provided herein.
5. Use **only blue or black ink pens** (no pencils or colored pens)

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**May 2026**



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**SECTIONA: ATTEMPT ALL QUESTIONS (55MARKS)**

**01.** Write the negation of the following" $(\exists x)[(p(x)\wedge q(x))\Rightarrow(\neg p(x))]$" **(3marks)**

A) $(\exists x)(p(x)\vee q(x))\Rightarrow p(x)$

B) $(\forall x)[p(x)\wedge q(x)]$

C) $(\forall x)[\neg p(x)\vee(\neg q(x)]\Rightarrow p(x)$

D) $(\exists x)(\neg p(x)\vee\wedge(\neg q(x)))$

**02.** Select the correct statement among the following **(2marks)**

A) The set of integers has the structure of commutative group with respect to addition operation

B) $\frac{\pi}{2}$ is a rational number

C) $2$ is a prime number

D) The set of irrational numbers is closed with respect to the multiplication operation

**03**. Simplify the following expression $E=\log_2\frac{8}{x}+\log_2\frac{x}{32}$ **(3marks)**

A) $E=2$

B) $E=0.25$

C) $E=-4$

D) $E=-2$

**04**. Perform the division of $f(x)=1+x^3$ by $g(x)=1+x$ **(3marks)**

A) $q(x)=1+x^2$

B) $q(x)=1-x^2$

C) $q(x)=1+x+x^2$

D) $q(x)=x^2-x+1$

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**05.** Answer by **TRUE (T)** or **FALSE (F)** in the following statements **(4marks)**

A) The quadratic equation has two repeated roots when discriminate is less than zero. **(1mark)**

B) The quadratic equation has no real roots when discriminate is equal to zero. **(1mark)**

C) The quadratic equation has two distinct real roots when discriminate is greater than zero. **(1mark)**

D) There is no x-intercept for the function $f(x) = -x^2 + x - 1$. **(1mark)**

**06**. Determine all values of $x$ the matrix $A = \begin{bmatrix} x & 1 & 1 \\ 1 & x & -1 \\ -1 & -1 & x \end{bmatrix}$ is singular **(5marks)**

A) $x \in \{1, -1, 2\}$

B) $x \in \{1, -1, 2\}$

C) $x \in \{1, -1, 0\}$

D) $x \in \{0, 1\}$

**07**. Determine the general solution of the equation $\cos x - \cos 3x = 0$ **(4marks)**

A) $x = k\pi, k \in \mathbb{Z}$

B) $x = (2k+1)\pi, k \in \mathbb{Z}$

C) $x = k\dfrac{\pi}{2}, k \in \mathbb{Z}$

D) $x = \dfrac{\pi}{3} + 2k\pi; k \in \mathbb{Z}$

**08**. Select the correct statement about the function $f(x) = \dfrac{x+1}{x-1}$ **(4marks)**

A) $f(x)$ has an oblique asymptote given by: $x + y = 0$

B) $f(x)$ has a vertical asymptote given by: $y = 1$

C) $f(x)$ has a horizontal asymptote given by: $x = -1$

D) The y-intercept of $f(x)$ is given by $(0, -1)$

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**09**. How many different ways can 5 people sit around a table **(3marks)**

A) 25

B) 120

C) 24

D) 5

**10.** Find the value of the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ **(4marks)**

A) $\infty$

B) Not defined

C) 0

D) 1

**11.** Evaluate the area of the region bounded by $f(x) = \ln x, x-$ axis and the line $x = e$) **(4marks)**

A) $e^{-1}$

B) $2e$

C) **1**

D) $2e^2$

**12.** In a partially destroyed laboratory record of an analysis of correlation data, the following results only are available. The regression lines equations are $x + 2y - 5 = 0; 2x + 3y - 8 = 0$ and the variance of the variable $x$ is12.Determine the mean values of $x, y$ and the variance of $y$. **(4marks)**

A) $(\bar{x}, \bar{y}, Var(y)) = (2,1,9)$

B) $(\bar{x}, \bar{y}, Var(y)) = (2,1,4)$

C) $(\bar{x}, \bar{y}, Var(y)) = (1,2,4)$

D) $(\bar{x}, \bar{y}, Var(y)) = (4,1,2)$

**13.** Three perfect coins are tossed simultaneously. Find the probability of getting at most one head . **(3marks)**



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A) 0.25

B) $\frac{5}{8}$

C) 0.5

D) $\frac{3}{8}$

**14.** Solve the differential equation**:** $y''+y=10e^{2x}$ **(5marks)**

A) $y=A\cos x+B\sin x+\frac{1}{3}\sin 2x$

B) $y=Ae^{-x}+Be^{x}+e^{2x}$

C) $y=A\cos x+B\sin x+2e^{2x}$

D) $y=Ae^{-x}+B\sin x+4e^{2x}$

**15**. We want to construct a box with a square base and we only have 10 square meters of material to use in construction of the box. If all the material is used in the construction process, Use the techniques of derivative and determine the maximum volume that the box can have. **(5marks)**

**SECTION B: CHOOSE ONLY THREE QUESTIONS (45MARKS)**

**16**. Given the function $f(x)=\frac{3x+1}{(x-1)(x^2+1)}$

a) Express $f(x)$ in partial fractions **(6marks)**

b) Evaluate the integral $\int \frac{3x+1}{(x-1)(x^2+1)}dx$ **(9marks)**

**17.** The point $P(x,y)$ represents a complex number $z=x+iy$ .Given that $|z+1|=2|z-1|$, defined in the set of complex numbers.

a) Find in terms of $x$ and $y$ the equation of locus of $P$ **(8marks)**

b) Describe the locus in words **(3marks)**

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c) Evaluate the area and length of the locus **(4 marks)**

18.a) Let $f(x)=\pi-x$ on the interval $[-\pi,\pi]$ with period $2\pi$. Find the Fourier series representation of $f(x)$ **(12marks)**

b) Let $f(x)=x$ on $[-L,L[$ be a periodic function with the period $P=2L$. Produce the graph on the interval $[-2L,2L[$. **(3marks)**

19. The continuous random variable X has probability density function given by: $f(x)=\begin{cases} kx(2-x), 0\le x\le 2 \\ 0, \text{elsewhere} \end{cases}$

a) Calculate the value of $k$ **(3 marks)**

b) Evaluate $p(0\le x\le 1)$ **(3 marks)**

c) Calculate the mode of the distribution **(2 marks)**

d) Find the variance $Var(x)$ **(7marks)**

**Q20.** a) The following results were obtained from Mathematics and Physics examinations:

| | Mathematics $(x)$ | Physics $(y)$ |
|---|---|---|
| Mean | 47.5 | 39.5 |
| Variance | 282.24 | 116.64 |

Given that the correlation coefficient is $r=0.95$,

i) Find equation of the regression line of $y$ in terms of $x$ **(6marks)**

ii) Estimate the value of $y$ for $x=35$ **(2marks)**

b) The regression lines of bivariate data of $x$ on $y$ and $y$ on $x$ are $20x+9y=107$ and $4x+5y+33=0$, respectively. Determine the marginal means and the correlation coefficient **(7marks)**

## END ASSESSMENT!



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